Program :

```
'''- Download a new dataset related to greenskilling or sustainability
  - Import in python file
  - Note all the generalisations and analysis about the dataset, note relations of each
feature/column to each other. (write it in text cells or comments)
  - if required do feature engineering
  - Handle null values according to your understanding of dataset
  - Used Seaborn library plots to visualise your dataset
  - Use scaling on numerical columns and pd.get_dummies / Label encoder on Categorical
columns
  - Give your insights and answer why you did what you did.'''
import pandas as p
from sklearn.preprocessing import *
import seaborn as s
import matplotlib.pyplot as plt
data = p.read_csv('data.csv')
print('Dataset Overview:')
data.info()
print('The person\'s age greater than 25 are:')
print(data[data['Age'] > 25])
print('Number of females in the dataset:')
print(data[data['Gender'] == 'Female'])
print('The peoples who belong to Ahmedabad city are:')
print(data[data['City'] == 'Ahmedabad'])
print('The number of null values in the dataset are:')
print(data.isnull().sum().sum())
if data.isnull().sum().sum() > 0:
    print('Handling null values by filling with mean for numerical columns and mode for
categorical columns.')
    for column in data.columns:
        if data[column].dtype == 'object':
            # Fill text columns with the most frequent value
            data[column] = data[column].fillna(data[column].mode()[0])
        else:
            # Fill numeric columns with the mean
            data[column] = data[column].fillna(data[column].mean())
print('Visualizing the dataset using Seaborn library:')
print('Heatmap of the dataset:')
corr = data.corr(numeric_only=True)
s.heatmap(corr, annot=True)
plt.show()
print('Pairplot of the dataset:')
s.pairplot(data)
plt.show()
print('Scaling numerical columns and encoding categorical columns:')
print(minmax_scale(data.select_dtypes(include=['float64', 'int64'])))
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print('Standard Scaled Data:')
print(StandardScaler().fit_transform(data.select_dtypes(include=['float64', 'int64'])))
print('Dummies for categorical columns:')
print(p.get_dummies(data, columns=['Gender', 'City']))
le = LabelEncoder()
data['Eco_Friendly'] = le.fit_transform(data['Eco_Friendly'])
print('Label Encoded Data:')
print(data)
print('Insights:')
print('1. This dataset provides information about individual\'s daily lifestyle use.')
print('2. The dataset contains both numerical and categorical features.')
print('3. Most of the people choose car as their mode of transport, which is not eco-friendly.')
print('4. The age of the people in the dataset is mostly between 20-30 years, which indicates
that younger people are more conscious about their lifestyle choices.')
```

Output:

```
Dataset Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 13 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Person_ID             15 non-null     int64
 1   Age                   15 non-null     int64
 2   Gender                15 non-null     object
 3   City                  15 non-null     object
 4   Transport_Mode        15 non-null     object
 5   Electricity_Usage     14 non-null     float64
 6   Water_Usage           14 non-null     float64
 7   Waste_Recycled        14 non-null     float64
 8   Solar_Panels          15 non-null     object
 9   Green_Certification   14 non-null     object
 10  Trees_Planted         14 non-null     float64
 11  Sustainability_Score  15 non-null     int64
 12  Eco_Friendly          15 non-null     object
dtypes: float64(4), int64(3), object(6)
memory usage: 1.7+ KB
The person's age greater than 25 are:
    Person_ID  ...  Eco_Friendly
3           4  ...           Yes
7           8  ...            No
9          10  ...            No
13         14  ...            No

[4 rows x 13 columns]
Number of females in the dataset:
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   Person_ID  ...  Eco_Friendly
0          1  ...           Yes
2          3  ...           Yes
4          5  ...           Yes
6          7  ...           Yes
8          9  ...           Yes
10        11  ...           Yes
12        13  ...           Yes
14        15  ...           Yes

[8 rows x 13 columns]
```

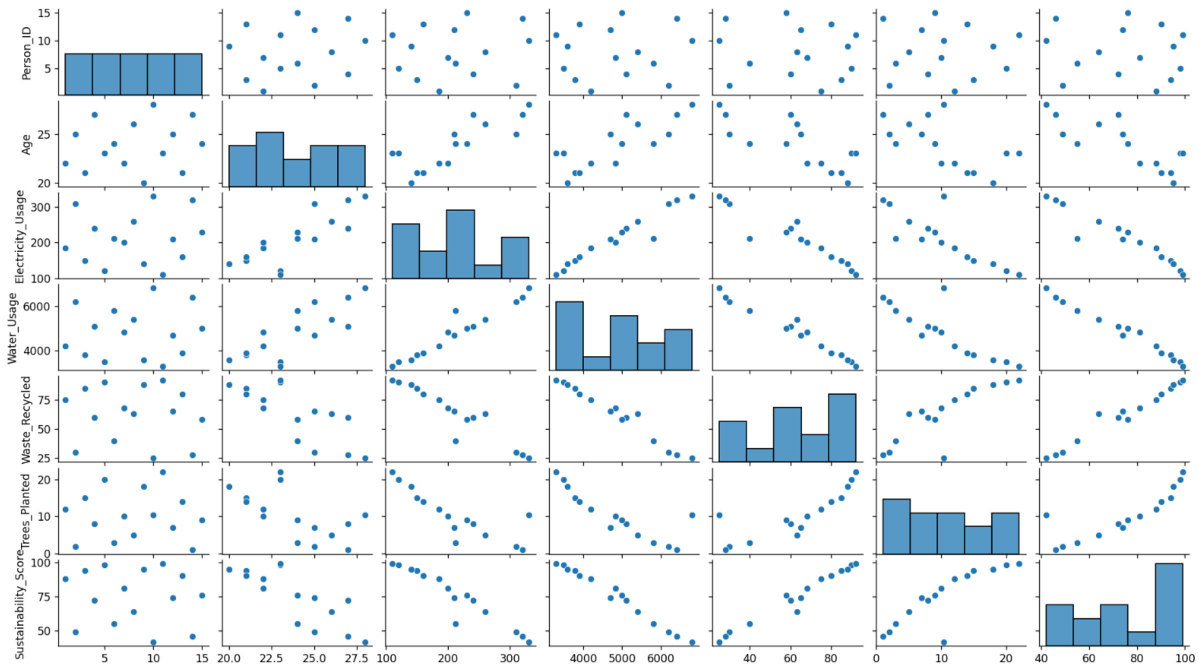
The peoples who belong to Ahmedabad city are:

```
   Person_ID  ...  Eco_Friendly
0          1  ...           Yes
4          5  ...           Yes
8          9  ...           Yes
12        13  ...           Yes

[4 rows x 13 columns]
```

The number of null values in the dataset are:
5
Handling null values by filling with mean for numerical columns and mode for categorical columns.
Visualizing the dataset using Seaborn library:
Heatmap of the dataset:

Pairplot of the dataset:



Scaling numerical columns and encoding categorical columns:

```
[[0.         0.25       0.34090909 0.25714286 0.74626866 0.52380952
  0.80701754]
 [0.07142857 0.625      0.90909091 0.82857143 0.07462687 0.04761905
  0.12280702]
 [0.14285714 0.125      0.18181818 0.14285714 0.89552239 0.66666667
  0.9122807 ]
 [0.21428571 0.875      0.59090909 0.51428571 0.52238806 0.33333333
  0.52631579]
 [0.28571429 0.375      0.04545455 0.05714286 0.97014925 0.9047619
  0.98245614]
 [0.35714286 0.5        0.46266234 0.71428571 0.2238806  0.0952381
  0.22807018]
 [0.42857143 0.25       0.40909091 0.43877551 0.64179104 0.42857143
  0.68421053]
 [0.5        0.75       0.68181818 0.6        0.56929638 0.19047619
  0.38596491]
 [0.57142857 0.         0.13636364 0.08571429 0.94029851 0.80952381
  0.92982456]
 [0.64285714 1.         1.         1.         0.         0.44897959
  0.        ]
 [0.71428571 0.375      0.         0.         1.         1.
  1.        ]
 [0.78571429 0.625      0.45454545 0.4        0.59701493 0.28571429
  0.56140351]
 [0.85714286 0.125      0.22727273 0.17142857 0.82089552 0.61904762
  0.84210526]
 [0.92857143 0.875      0.95454545 0.88571429 0.04477612 0.
  0.07017544]
```

```
 [1.         0.5        0.54545455 0.48571429 0.49253731 0.38095238
  0.59649123]]
Standard Scaled Data:
[[-1.62018517e+00 -7.90065707e-01 -3.91877843e-01 -5.87682188e-01
   5.31626198e-01  2.51968603e-01  6.90035503e-01]
 [-1.38873015e+00  4.79682751e-01  1.43688542e+00  1.26120559e+00
  -1.48599130e+00 -1.35146796e+00 -1.35905470e+00]
 [-1.15727512e+00 -1.21331519e+00 -9.03931557e-01 -9.57459745e-01
   9.79985641e-01  7.32999572e-01  1.00528015e+00]
 [-9.25820100e-01  1.32618172e+00  4.12777994e-01  2.44317314e-01
  -1.40912968e-01 -3.89406023e-01 -1.50616886e-01]
 [-6.94365075e-01 -3.66816221e-01 -1.34283474e+00 -1.23479291e+00
   1.20416536e+00  1.53471785e+00  1.21544325e+00]
 [-4.62910050e-01  5.64332648e-02  0.00000000e+00  8.91428038e-01
  -1.03763186e+00 -1.19112430e+00 -1.04381005e+00]
 [-2.31455025e-01 -7.90065707e-01 -1.72426251e-01 -8.40776821e-16
   2.17774587e-01 -6.87187099e-02  3.22250083e-01]
 [ 0.00000000e+00  9.02932237e-01  7.05380117e-01  5.21650482e-01
   3.18578546e-16 -8.70436992e-01 -5.70943081e-01]
 [ 2.31455025e-01 -1.63656468e+00 -1.05023262e+00 -1.14234852e+00
   1.11449347e+00  1.21403054e+00  1.05782092e+00]
 [ 4.62910050e-01  1.74943121e+00  1.72948754e+00  1.81587193e+00
  -1.71017102e+00  0.00000000e+00 -1.72684012e+00]
 [ 6.94365075e-01 -3.66816221e-01 -1.48913580e+00 -1.41968169e+00
   1.29383725e+00  1.85540517e+00  1.26798402e+00]
 [ 9.25820100e-01  4.79682751e-01 -2.61251895e-02 -1.25460242e-01
   8.32667538e-02 -5.49749679e-01 -4.55353378e-02]
 [ 1.15727512e+00 -1.21331519e+00 -7.57630496e-01 -8.65015356e-01
   7.55805920e-01  5.72655916e-01  7.95117051e-01]
 [ 1.38873015e+00  1.32618172e+00  1.58318648e+00  1.44609437e+00
  -1.57566319e+00 -1.51181162e+00 -1.51667702e+00]
 [ 1.62018517e+00  5.64332648e-02  2.66476933e-01  1.51872925e-01
  -2.30584857e-01 -2.29062366e-01  5.95462109e-02]]
Dummies for categorical columns:
    Person_ID  Age  ... City_Surat  City_Vadodara
0           1   22  ...      False          False
1           2   25  ...       True          False
2           3   21  ...      False           True
3           4   27  ...      False          False
4           5   23  ...      False          False
5           6   24  ...       True          False
6           7   22  ...      False          False
7           8   26  ...      False           True
8           9   20  ...      False          False
9          10   28  ...       True          False
10         11   23  ...      False          False
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11         12   25  ...      False          True
12         13   21  ...      False         False
13         14   27  ...       True         False
14         15   24  ...      False         False

[15 rows x 17 columns]
Label Encoded Data:
    Person_ID  ...  Eco_Friendly
0           1  ...             1
1           2  ...             0
2           3  ...             1
3           4  ...             1
4           5  ...             1
5           6  ...             0
6           7  ...             1
7           8  ...             0
8           9  ...             1
9          10  ...             0
10         11  ...             1
11         12  ...             1
12         13  ...             1
13         14  ...             0
14         15  ...             1

[15 rows x 13 columns]
```

Insights:

1. This dataset provides information about individual's daily lifestyle use.
2. The dataset contains both numerical and categorical features.
3. Most of the people choose car as their mode of transport, which is not eco-friendly.
4. The age of the people in the dataset is mostly between 20-30 years, which indicates that younger people are more conscious about their lifestyle choices.